

# THE Aesthetic BRAIN

*How does beauty move you and art sculpt minds according to neuroaesthetics?*

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**H**ave you ever heard the saying that beauty is in the eye of the beholder? This saying means that a sight one person deems visually appealing may not have the same effect on another. Aesthetics and beauty are highly personal values, shaped

by someone's experience and cultural background. However, some of this quality is universal, shaped by the brain's architecture and interpretation of a visual.

Neuroaesthetics is a topic centred on the brain's perception and response to art and beauty. It combines neuroscience and psychology, and explores why we

are drawn to some imagery more than others. What happens in our brains when we browse an art gallery, scan a dramatic landscape or subconsciously rate a person's looks? By using neuroimaging



tools, such as fMRI (functional magnetic resonance imaging), scientists can analyse the inner workings of the brain when face-to-face with beauty.

### **Beauty on the brain**

One of the main brain regions shown to alight on scans when a person finds something aesthetically pleasing is the orbitofrontal cortex (OFC). This region analyses stimuli to decide its emotional value. When a person perceives beauty, the OFC activity increases. If there is increased activity in the OFC, it is more likely that the brain is linking stimuli with feelings of positivity and desire.

Beauty also impacts the ventral striatum of the brain. This area includes the nucleus accumbens, which releases a neurotransmitter called dopamine. Neurotransmitters are chemical messengers that regulate emotions and perceptions. Dopamine, in particular, is associated with pleasure and reinforcement. Finally, the anterior cingulate cortex plays a role in the brain's emotional regulation. When viewing beauty, this region of the brain combines cognitive, visual and emotional information to establish a feeling and to process what this reaction to the stimuli means.

Sometimes, people are criticised for their bias towards people based on their looks, giving those considered 'better looking' more advantages and opportunities. According to science, there could be a biological reason for people favouring visually pleasing stimuli. The

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dopamine that is released in the ventral striatum is like a drug. Dopamine helps the brain to cement connections between a feeling of pleasure and an act. In the case of neuroaesthetics, the act is seeing something beautiful - and this causes the brain to seek attractive sights over those that don't spark the same response.

Most of the brain's workings are deep-rooted in survival. But, how can something that seems so trivial and superficial be a product of evolution? The leading theory as to why beauty is a factor to the human brain is that the human body is wired to enjoy shapes and forms that are present in nature, as these patterns have helped our ancestors to survive. For example, a symmetrical face might indicate good health and strong genes. Ancestors who favoured these qualities were more likely to reproduce with healthy mates and grow the population. This quality stretched far beyond other people. Plants that grow in fractal patterns - meaning their complex shape is made up of a repeating pattern or process - are considered safe to eat by the brain. This includes foods like broccoli and cauliflower that display a shape formed by a repeating and natural

growth process. Heavy metals and other pollutants can contaminate plants and prevent this natural growth. By favouring these aesthetic patterns in food, the brain is protecting the body from contaminants.

### **Is beauty subjective?**

The impact of evolutionary preferences means that most of the population share core values in the brain's interpretation of beauty. Something is more likely to be aesthetically pleasing if it is symmetrical, balanced and proportional. Universally, these traits are agreed to be the most attractive in a person, and studies show that even babies act more favourably towards people who adults judge to be more attractive. These biases aren't learned, but ingrained in you from birth.

Symmetry, balance and proportion are important in aesthetics across the board, not just in human features. The brain is able to process symmetrical patterns more efficiently than asymmetrical ones, which require more cognitive effort.

Beauty isn't confined to order, though. Contrast and asymmetry ignite curiosity in the brain and build dramatic tension. This activates neural circuits in the brain »





## HOW CAN ART REWIRE THE BRAIN?

Art can be beneficial in restructuring the brain in those with some neurological conditions. During forms of art-based therapy, the brain's emotional and aesthetic systems are engaged. Artistic activities, such as painting, music and dance, can stimulate multiple neural pathways in the brain simultaneously, improving communication between sensory, motor and cognitive regions. Such activities can benefit fine motor control and attention, and studies show they help to reduce depression and anxiety, due to art releasing dopamine and activating the brain's reward circuits.

The beauty of art can offer far more than general aesthetic pleasure. Viewing collections of colour and patterns that go well together can calm an overactive stress response, while creating an artistic piece activates dormant neural pathways in an individual's brain, including those linked to coordination and perception. People are often able to release feelings into artwork as a way to express themselves and share feelings that may be more difficult to talk about in therapy. Colour and imagery are closely linked to mood, and engaging with these elements has great power in restoring and altering a person's emotions.

for emotion and attention. We crave uniqueness in art and, if all imagery followed strict rules, creative freedom would be lost and the brain's reaction to art would be too repetitive. Instead, the organ thrives off novel stimuli. This activates the release of dopamine and is what draws our fascination to contrasting patterns in artwork and surreal imagery.

### **How do we perceive art?**

When you lay your eyes on a carefully constructed canvas, combining and balancing textures, colours, shapes and

tones, your senses and brain respond to each other, igniting a reaction in the body. Aesthetics literally alter your brain chemistry, and this is why humans value art and design so highly. Humans continuously organise their living spaces, paint the walls of their homes with their favourite colours, and are particular about the attire they wear. These aren't trivial factors of a person's daily routine or home environment, but a way to shape their wellbeing. Certain colours will bring happiness to an individual, and an orderly home can relieve stress. Sight is one of

the main senses you use to perceive the world, interpret expressions and evaluate surroundings. They help to tell the brain when to panic, when to relax and what to deduce from intricate details. This is a natural response in engaging with the world around you, so when an artist channels all known artistic theory to a canvas, with the goal to initiate an emotional reaction, the results can be powerful.

The fundamental features of a painting, like line, colour, shape and movement, are processed by the visual cortex in the brain. This information is then sent to higher-order association areas, which analyse the meaningful patterns that these shapes make. This step is essential to see a person's face in the combination of colour applied to a portrait painting, for example.

All of our interpretations of artwork are shaped by context, memory and expectation. This is why people have different responses to the same piece of art. In some people, paintings can evoke strong reactions as they associate objects, shapes or colours with a traumatic experience, positive memory or current emotional state. If the elements of an image hold personal significance, a person is more likely to find beauty - or another strong reaction - in the aesthetic. Finally, the expectation element of artwork comes from the brain's tendency to predict and interpret. It is constantly forming mental models of the world, in order to predict stimuli and make rapid responses for survival purposes. Because of this, your brain won't just see what has been painted. It will automatically search for hidden images, engage with stimulating regions and attempt to make sense of the artwork. The results of this process will alter a person's opinion of the image.

### **Emotional, reward and sensorimotor systems**

Beauty evokes emotion by engaging the brain's emotional, reward and sensory-motor systems at the same time. Art activates the limbic system, which is a group of interconnected brain structures working to regulate your emotions and behaviours. The amygdala and hippocampus are the specific regions responsible for emotional memory and emotional response. This reaction can tie emotions to art, reminding people of other memories and bringing powerful emotions to the surface.

Endorphins are also released when the brain engages with art. More of these feel-good hormones flood the body when experiencing something you perceive as beautiful. This reward system makes a person seek out beauty subconsciously. Finally



distributed throughout the brain, nerves and spinal cord. It processes movement and touch, so it becomes a factor in art forms such as dance and sculpture. The additional senses in tactile art can enable another level of emotional response. Sensorimotor interaction deepens the definition of beauty. It doesn't have to be a visual experience, but a feeling your body experiences. Research shows that the more sensorimotor interaction children have with an object, the higher they rate it in terms of its beauty.

### **Mirroring art**

One fascinating factor of neuroaesthetics is the human ability to mirror artwork. You may view looking at art as a passive event, in which you take in the visual image only. However, viewing art engages the mirror neuron system. This is a network of neurons that are activated when we see an act, whether that be in real time or portrayed in an image. The neurons are located in the premotor cortex and inferior parietal lobe of the brain, and are essential in conveying empathy and aligning emotion.

In essence, mirror neurons cause a viewer to actively engage in the artwork they are looking at. If looking at a painting that depicts grief, the brain mimics the feelings that the artist is portraying in the observer. Art does much more than display emotion. It can be so moving because it actively transmits the emotion into people who see it. As well as replicating a visual scene, it replicates the feeling of the frozen moment in time.

### **The future of neuroaesthetics**

Scientists are continuing to discover the power of aesthetics and its impact on the human brain. When your eyes take in the world, what you see can significantly

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## THE BIOPHILIA HYPOTHESIS

Have you ever looked out over a landscape and been mesmerised by its stunning sight? In general, human senses perceive nature to be soothing and visually pleasing. The biophilia hypothesis, first proposed by biologist Edward Wilson, suggests that humans feel connected to nature and find ways to tune in to other living systems.

This hypothesis is based on the idea that our brains are naturally attuned to the patterns and physical forms of nature, and that this connection has formed over millions of years as we have evolved. For example, greenery and water have supported the survival of our species, and so, over time, the human brain has learned to connect these sights with positive emotion.

Being in the wild therefore reduces stress and benefits the brain.

Multiple scientific studies have supported elements of this hypothesis. For example, natural visual stimuli has shown to lower cortisol levels and activate parts of the brain that provide pleasure and empathy. The same areas of the brain that are triggered by art and beauty are also energised by natural forms.

shape your experience of it. With new technology emerging, such as brain-computer interfaces, neuroaesthetics can be better measured to decode the effect of different stimuli on the brain. This could lead to more qualitative and personalised data of each individual's aesthetic preferences and how they

respond to colour, shapes, music and other forms of art. Humans have created and enjoyed art for tens of thousands of years, producing cave paintings and establishing our boundaries of beauty. But, people are only just beginning to understand the true influence of art and aesthetics, and how it completely captivates minds and bodies. ■

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